

Architecture, Scope, and Goals of the Multidisciplinary DPW-8/AePW-4 Workshop

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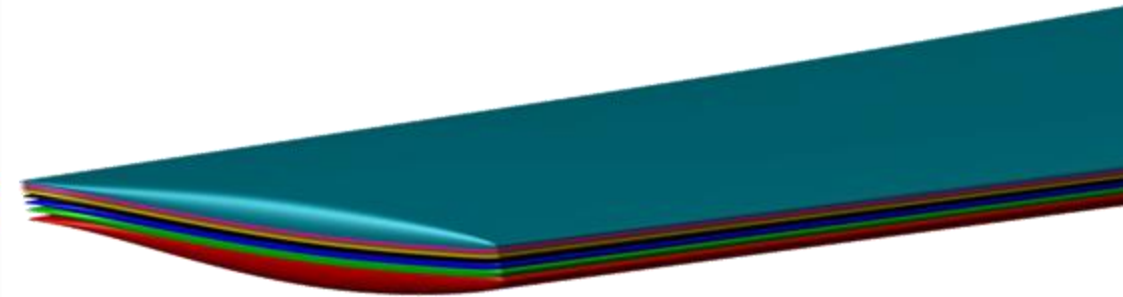
Aeroelasticity Branch
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High-Speed Aerodynamics
Boeing Commercial Airplanes Flight Sciences



- **Next generation of computational goals is highly multidisciplinary**
 - CFD 2030 highlights edge-of-the-envelope technology needs
 - Knowledge exists in various fields, but lessons learned remain isolated
 - Identifies the need for collaboration via symbiotic partnerships
- **Workshop Collaboration Working Group**
 - Chartered by Applied Aero TC (2022)
 - Assembled four individuals for a key interdisciplinary organizing committee
 - Panels, presentations, and meetings provided a forum to continue conversations
 - Hallway conversations led to DPW/AePW collaboration
- **SciTech 2024 kickoff meeting**



Experimentally determined
lofts from DPW-VII

- **Strengths in communities**

- DPW: high-fidelity, transonic CFD including steady and unsteady schemes
- AePW: multifidelity steady and unsteady aeroelastic computations, including structural and aerodynamic nonlinearities

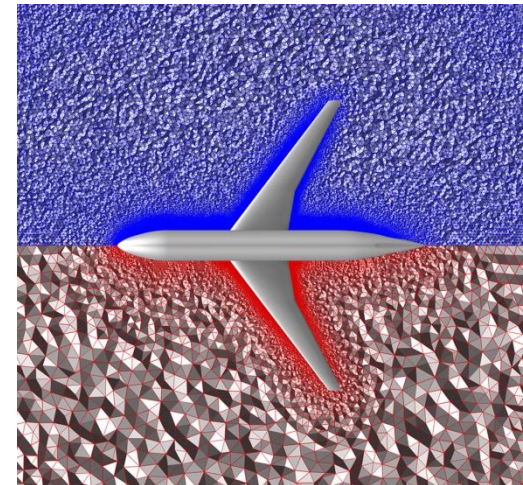
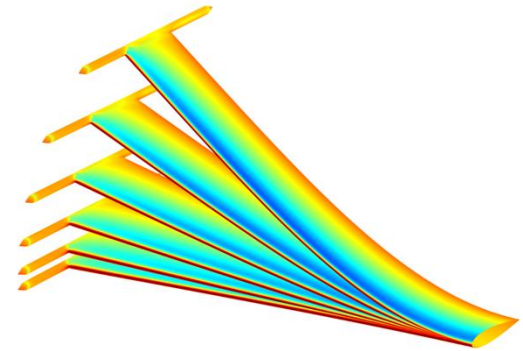
- **Opportunities for growth**

- DPW: increased fidelity aeroelastic methods
- AePW: increased geometric complexity and aerodynamic models

- **Workshop scope**

- Mature each community's state-of-the art
- Enable multidisciplinary technical advances to improve the cutting edge and foster cross-pollination
- Develop multidisciplinary workshop model

- **Selected a working-group model**



Workshop Leadership Global Presence



Source: OpenStreetMap
Open source, subject to Open Database License

- **Interdisciplinary**

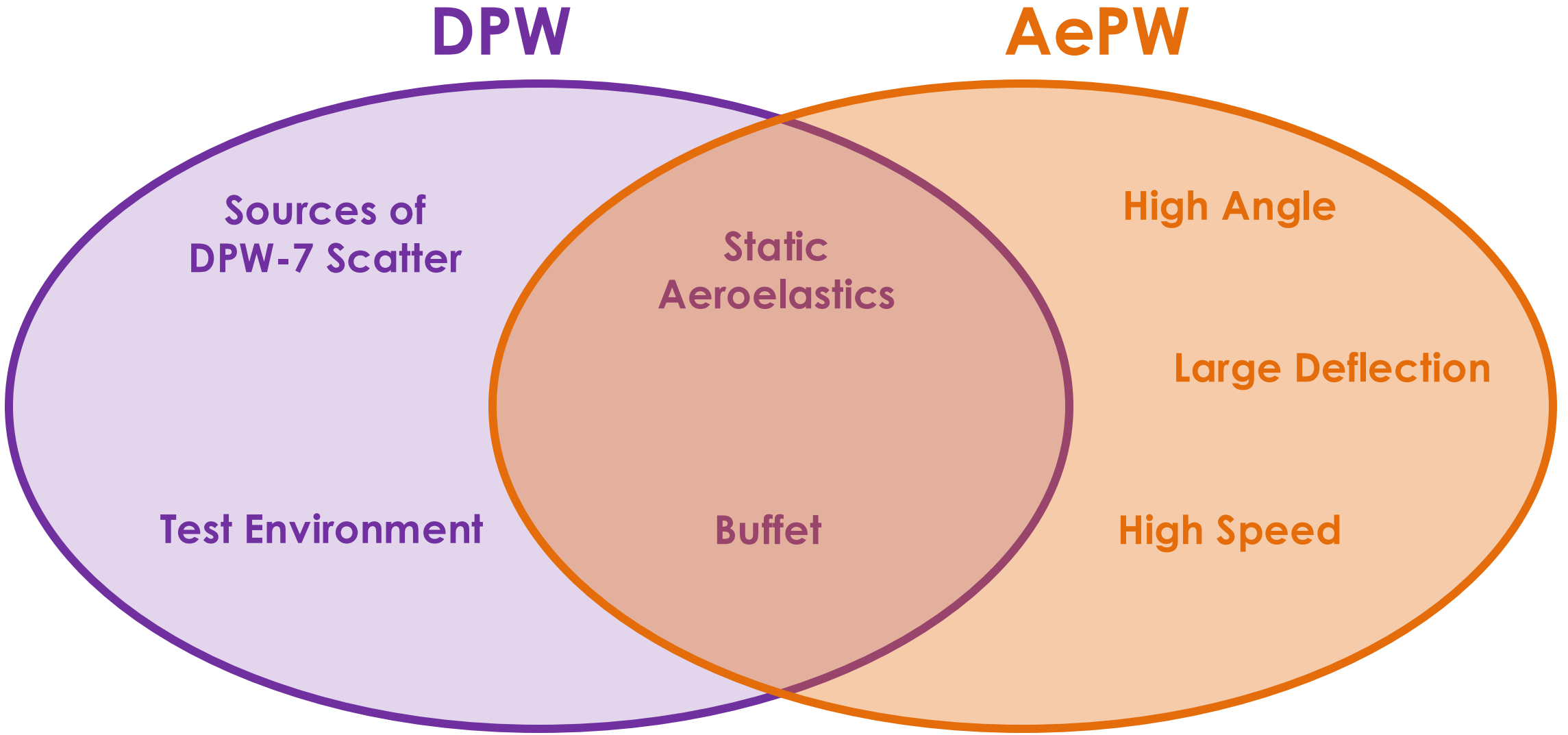
Pawel Chwalowski, US
Brent Pomeroy, US
Ben Rider, US
Bret Stanford, US

- **AePW**

Kirk Brouwer, US
Carlos Cesnik, US
Pawel Chwalowski, US
Adam Jirasek, US
Jeff Ouellette, US
Rafael Palacios, UK
Daniella Raveh, Israel
Markus Ritter, Germany
Walt Silva, US
Bret Stanford, US

- **DPW**

Hadar Ben-Gida, Israel
Seth Kelly, US
Stefan Keye, Germany
Mitsuhiro Murayama, Japan
Raj Nangia, UK
Ilias Petropoulos, France
Brent Pomeroy, US
Andrea Sansica, Japan
Ben Rider, US
Melissa Rivers, US
Ed Tinoco, US
Chris Toomer, UK
John Vassberg, US

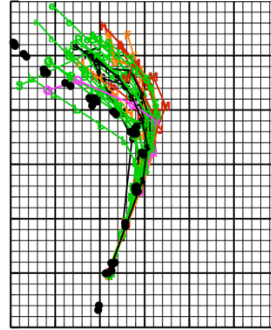


- **Deconflicted seven working groups recurring meeting time**
 - Settled on a least-bad meeting time for all time zones
 - Bi-weekly or monthly cadence
 - Some groups alternate time every-other meeting
- **Working group leadership entrusted with scoping group's efforts**
- **Buffet Working Group further segmented**
 - Subgroups selected by scheme
 - Three groups meet during the month, one asynchronous (email) group

Community-Centric Working Group Goals

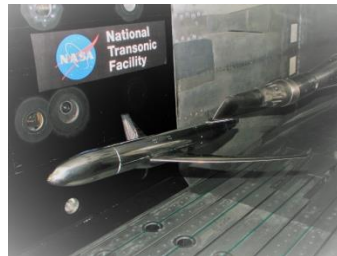
- **Sources of DPW-7 Scatter**

- Increase understanding and quantify expectations for comparisons between CFD and measured wind tunnel “truth”



- **Test Environment**

- Resolve differences between free-air CFD and the test environment



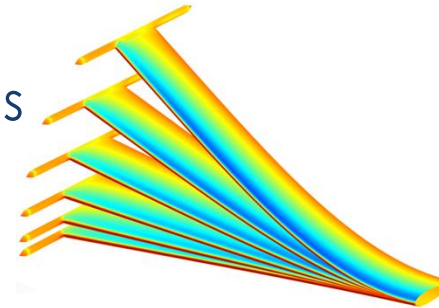
- **High Angle**

- Improve flutter predictions in the transonic region using a variety of CFD models



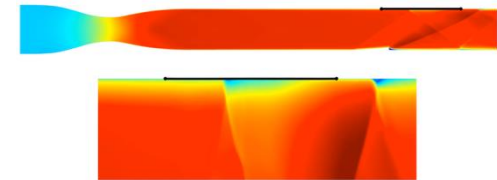
- **Large Deflection**

- Mature predictions for low-speed deformations with nonlinear stiffness



- **High Speed**

- Advance predictions of supersonic fluid structure interactions



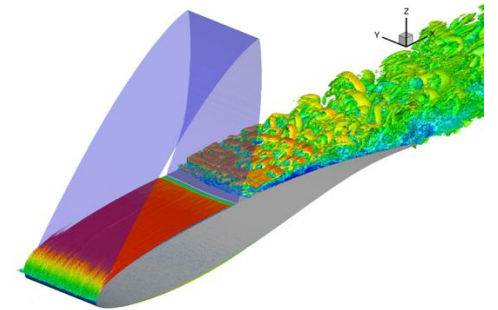
- **Static Deformation**

- Determine accuracy and uncertainty of transonic, center-of-the-envelope wing calculations



- **Buffet**

- Establish state of the art and identify opportunities for improvement of transonic, high-alpha unsteady CFD and unsteady coupled FSI



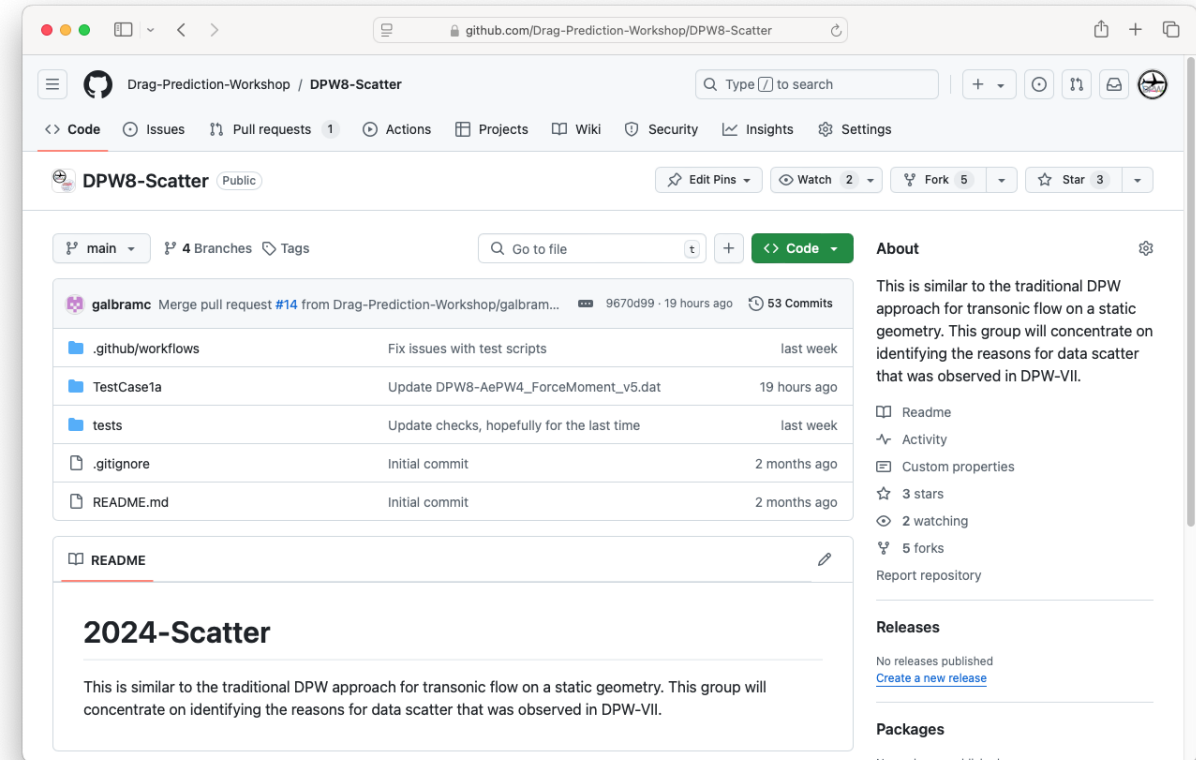
- **Establish key questions to be answered**
 - Three working groups continued from previous workshops
 - Two working groups were direct fallout from workshop lessons learned
 - Two hybrid working groups are new
- **Engage students and low-computational-capacity contributors**
 - Coarsest simulations scoped to run on a laptop
 - Short-term Tecplot licenses available to students

- **Two-day, pre-conference workshop at AVIATION '26**
- **First day**
 - Community centric in two separate rooms
 - Technical lessons learned
 - Future plans
- **Second day**
 - Everyone together
 - Hybrid groups
 - Workshop lessons learned
 - Future plans
- **Potential plans**
 - Miniature expo
 - Student poster session

- **Organizing committee formulation is key**
 - Interdisciplinary Organizing Committee
 - DPW Organizing Committee
 - AePW Organizing Committee
 - Working Group Organizing Committees
- **Clear, consistent communication is key**
 - Interdisciplinary OC members also attending community-centric meetings
 - Development of the key questions is encouraged to support a central theme
 - Consistent touch-base communication and meetings

- **Quarterly workshop-wide tagups**
 - Workshop overview provides insight into all working groups
 - Technical scope and updates from all working groups
 - Provides opportunity for follow-on Q&A
 - Encourages individuals to engage with opposite-community working groups
- **Hybrid working groups**
 - Empower working group leaders to develop key questions under workshop scope
 - Collaborative effort between communities to formulate the group's goals
 - Identify strengths and weakness of community simulation methods and establish opportunities for multidisciplinary advancements
 - Strong presence from both DPW and AePW communities

- **One stop shop for four working groups data submission**
 - Improves version control
 - Ensures all committee members are analyzing the most recent data
 - Responsibility lies with participants to ensure data are up to date
- **Public and outward-facing**
- **Automatic checks ensure files are in the appropriate format**



- **Multiple special sessions at AVIATION highlight group progress to date**
 - Committee presentations
 - High-Speed AePW Working Group
 - Buffet DPW/AePW Working Group
 - High-Angle and Large Deflection AePW Working Groups
- **Mini workshop scheduled for SciTech (as a panel session, open to all)**
- **Full workshop at AVIATION next year**

Questions?



<https://www.aiaa-dpw.org>



<https://nescacademy.nasa.gov/workshops/AePW4/public>